Title : Experiment With Regularization Techniques

1. Aim of the Experiment

To study and compare the effect of regularization techniques on a neural network by (1) training an MNIST classification model without regularization, (2) training the model using L2 regularization, (3) adding Dropout and comparing its performance, and (4) visualizing and analyzing training and validation loss and accuracy.

2. Theory

2.1 Regularization

In deep learning, a model may perform very well on training data but poorly on unseen data. This problem is known as overfitting. Regularization is a technique used to reduce overfitting and improve the model's ability to generalize to new data. In this experiment, three models are compared: Model 1 without regularization, Model 2 with L2 regularization, and Model 3 with Dropout.

2.2 Model Without Regularization

The first model consists of a 28×28 input image, a Flatten layer, a Dense layer with 128 neurons using ReLU activation, and an output Dense layer with 10 neurons using Softmax activation. The model learns the weights directly without imposing a regularization constraint.

2.3 L2 Regularization

L2 regularization adds a penalty to the loss function based on the squared values of the weights.

$$L_{total} = L_{data} + \lambda \sum w_i^2$$

where L_data is the original classification loss, w_i represents model weights, and λ is the regularization parameter. In this experiment λ = 0.001. L2 regularization discourages excessively large weights and can improve generalization.

2.4 Dropout Regularization

Dropout randomly deactivates a fraction of neurons during training. In this experiment, the dropout rate is 0.5, so approximately 50% of the neurons are randomly ignored during each training iteration. This reduces dependence on particular neurons and can help reduce overfitting. Dropout is applied during training and is automatically disabled during testing/inference.

3. Dataset Used

Parameter	Description
Dataset	MNIST
Image size	28 × 28 pixels
Number of classes	10
Original pixel values	0–255
Normalized values	0–1
Training samples	60,000
Test samples	10,000
Validation split	20% of training data

The pixel values are normalized using x_normalized = x / 255, converting the values from 0–255 to the range 0–1 for more stable training.

4. Architecture of the Models

All three models use the same basic neural network architecture:

MNIST 28×28 → Flatten (784) → Dense (128, ReLU) → Dense (10, Softmax)

Model 1 – Without Regularization: 28×28 → Flatten → Dense(128, ReLU) → Dense(10, Softmax)

Model 2 – L2 Regularization: 28×28 → Flatten → Dense(128, ReLU + L2) → Dense(10, Softmax)

Model 3 – Dropout: 28×28 → Flatten → Dense(128, ReLU) → Dropout(0.5) → Dense(10, Softmax)

5. Algorithm

1. Import TensorFlow, Matplotlib, Keras layers, Adam optimizer and the L2 regularizer.

2. Load the MNIST training and testing datasets.

3. Normalize all image pixel values by dividing them by 255.

4. Create Model 1 with Flatten, Dense(128, ReLU), and Dense(10, Softmax). Compile it using Adam, sparse categorical cross-entropy, and accuracy.

5. Train Model 1 for 10 epochs using 20% of the training data for validation.

6. Create Model 2 using the same architecture and add L2 regularization with $\lambda = 0.001$ to the hidden Dense layer.

7. Train Model 2 for 10 epochs and store its training history.

8. Create Model 3 by adding Dropout(0.5) after the hidden Dense layer.

9. Train Model 3 for 10 epochs and store its training history.

10. Evaluate all three models using the MNIST test dataset and record test loss and accuracy.

11. Plot training and validation accuracy for all three models.

12. Plot training and validation loss for all three models.

13. Compare the curves and test results to study the effect of regularization.

6. Flowchart

START

↓

Import Libraries

↓

Load MNIST Dataset

↓

Normalize Images (Pixel / 255)

↓

Model 1: Without Regularization

↓

Train for 10 Epochs (Validation = 20%)

↓

Model 2: L2 Regularization ($\lambda = 0.001$)

↓

Train for 10 Epochs

↓

Model 3: Dropout (0.5)

↓

Train for 10 Epochs

↓

Evaluate All Models: Test Loss & Accuracy

↓

Plot Training/Validation Accuracy

↓

Plot Training/Validation Loss

↓

Compare & Analyze

↓

END

7. Analysis and Discussion

The three models are trained using the same MNIST dataset, network architecture, optimizer, number of epochs, and validation split. The main difference is the regularization technique.

Model 1 – Without Regularization: The model can achieve high training accuracy because it is free to fit the training data. If training accuracy continues to increase while validation accuracy stops improving, or if validation loss starts increasing, this indicates possible overfitting.

Model 2 – L2 Regularization: L2 adds a penalty for large weights. Training accuracy may be slightly lower than the unregularized model, but the gap between training and validation performance can be reduced. The value 0.001 controls the strength of the penalty; an excessively large value may cause underfitting.

Model 3 – Dropout: With a dropout rate of 0.5, neurons are randomly deactivated during training. Training accuracy may be lower and training loss may be higher during training, while validation performance can become more stable. Dropout reduces the model's dependence on particular neurons.

Training accuracy alone should not be used to judge the effect of regularization. Training curves, validation curves, and independent test performance should be considered together.

8. Observation Table

Model	Regularization	Test Accuracy	Observation
Model 1	None	_____ %	Baseline model
Model 2	L2, $\lambda = 0.001$	_____ %	Controls large weights
Model 3	Dropout = 0.5	_____ %	Reduces neuron dependency

Graph Analysis: The accuracy graph should be examined for the difference between training and validation accuracy, improvement or stabilization of validation accuracy, and signs of overfitting. The loss graph should be examined for reduction in training loss, behavior of validation loss, and the gap between training and validation loss.

9. Conclusion

The experiment demonstrates the importance of regularization in neural network training. Three MNIST classification models were developed: one without regularization, one using L2 regularization, and one using Dropout. L2 regularization controls the magnitude of model weights by adding a penalty to the loss function, whereas Dropout randomly deactivates neurons during training. Training and validation accuracy and loss curves provide a visual way to study overfitting and generalization. Thus, L2 regularization and Dropout are useful techniques for controlling overfitting, and their effect should be evaluated using validation behavior and independent test performance rather than training accuracy alone.